Berezin reconstruction for the disk-with-cone truncated spectral triple, and the apex backbone of the cigar-tip entropy

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Abstract

The metric-spectral-triple convergence program underlying the GeoVac framework [9, 11, 13, 15] has so far treated compact groups and homogeneous spaces, where the central Fejér kernel of Peter–Weyl supplies the Berezin reconstruction map. We treat the first non-group, non-homogeneous case: the two-dimensional *disk-with-cone* $\mathbb{D}_{R,\alpha}$, a flat disk of radius R with a conical defect of apex multiple α at its centre and a Dirichlet boundary. This is the spatial factor of the near-horizon cigar whose truncated heat trace carries the Bekenstein–Hawking–type tip entropy of the companion Paper 51. We construct a *unital Markov–Cesàro Berezin* reconstruction $\mathcal{B}_\Lambda$ on $\mathbb{D}_{R,\alpha}$ and prove that, on apex-localised observables, it has exact unitality (by the Markov normalisation), while Cesàro positivity *also* holds on the finite disk at order $s \geq 2$, the remaining property — a convergent approximate-identity interior rate — is *obstructed on the finite disk* by the Dirichlet boundary (see below) and is recovered on the boundaryless plane. Together with the propagation-number-two operator system and the gradient-norm Lipschitz seminorm, this yields the propinquity backbone for the cigar-tip entropy of Paper 51 at the local apex observable, upgrading its Lemma on geometric faithfulness from cited to proved for that purpose. The construction is verified numerically: disk unitality is exact and the Cesàro construction of Definition 3.1 at order $s \geq 2$ *does* preserve positivity on the finite disk (minimum eigenvalue > 0 across the Λ sweep), but it does *not* achieve a convergent interior rate (the disk drivers give $\|\mathcal{B}_\Lambda(f) - P_\Lambda M_f P_\Lambda\|/\|\nabla f\|_\infty \sim \Lambda^{+0.07}$, i.e. no decay) — this is the Dirichlet-boundary obstruction, not a tail effect. (The minimum eigenvalue reaches ≈ -0.13 only for the truncated *heat* weight $e^{-t\lambda}$, which is Gibbs-non-positive; the Cesàro weight restores positivity but not convergence, so the genuine obstruction is the non-decaying rate — weight-robust — not positivity.) The *boundaryless plane* reconstruction (below) is what genuinely converges: the 2D Bochner–Riesz Berezin is positivity-preserving at order $s \geq \frac{1}{2}$ (classical) and reconstructs every Schwartz observable at algebraic rate $\Lambda^{-0.6}$ to $\Lambda^{-0.9}$ (test-function dependent). The only obstruction — observables with content at the Dirichlet boundary $\rho = R$, where the eigenfunctions vanish and the Markov ratio becomes a 0/0 limit independent of the observable — is an artifact of the finite-radius compactification, not of the function's decay, and is *not* removed by the naive $R \to \infty$ limit of the disk. The cure is the boundaryless carrier itself: on the plane $\mathbb{R}^2$ (the de-compactified cone), the band-limited Bochner–Riesz Berezin reconstructs every Schwartz observable with all four properties at an algebraic rate. Assembling Latrémolière's pointed/proper propinquity from these ingredients — properness by Arzelà–Ascoli, reach the reconstruction rate, height the gradient-non-expansive Bochner–Riesz distortion — establishes plane (hence de-compactified disk-with-cone) convergence at the qualitative-rate level, in van Suijlekom's state-space Gromov–Hausdorff sense; the strictly stronger *Latrémolière* propinquity carries the same named pointed/proper bookkeeping residual that the compact companion Papers 38/40 carry (the self-contained sharp-constant bound is the open residual). This is the metric-level analogue of the non-compact limit that Paper 47 closes at the spectral level.

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1 Introduction

1.1 Context

Latrémolière's Gromov–Hausdorff propinquity [1, 2] makes precise the convergence of a sequence of truncated metric spectral triples to a limiting (commutative or noncommutative) metric geometry. In the spectral-truncation setting of Connes and van Suijlekom [4], where the truncated operator system is $\mathcal{O}_n = P_n C(M) P_n$ for a spectral projection P_n, the GeoVac series of companion papers has established van Suijlekom state-space Gromov–Hausdorff convergence for the round three-sphere as SU(2) [9], for tensor products of such factors [10], for all simple compact Lie groups with a universal rate constant $4/\pi$ [11], and — in the Lorentzian/Krein setting — has probed the signature extension in Papers [12, 15] (the genuinely Lorentzian-metric propinquity *degenerates*: Paper 45's K^+ annihilation theorem, so the surviving statement there is the signature-agnostic product-carrier convergence, not a Lorentzian convergence theorem). In each of the established (Riemannian) cases the reconstruction map (the L4 lemma in the five-lemma scheme of Paper 38) is built from the *central Fejér kernel* of Peter–Weyl analysis, which exists because the carrier is a group or a homogeneous space and is positive by Fejér's classical theorem.

The present paper treats the first carrier that is neither: the two-dimensional flat *disk* $\mathbb{D}_R$, optionally with a conical defect at the centre (the *disk-with-cone* $\mathbb{D}_{R,\alpha}$), and a Dirichlet boundary at $\rho = R$. Two features make it genuinely new. First, the disk is not a group, so there is no central Fejér kernel; a reconstruction map must be built by hand. Second, and decisively, the disk is a manifold *with boundary*, which no previous carrier in the series was — and the boundary turns out to be the only obstruction to a full propinquity statement.

1.2 Motivation: the cigar-tip entropy

The disk-with-cone is the spatial factor of the Euclidean near-horizon cigar geometry $\mathbb{D}_{R,\alpha} \times S^2(r_h)$ at constant warp. The companion Paper 51 studies the replica derivative of the truncated heat trace on this geometry: the conical defect α lives entirely in the disk via the spinor azimuthal momentum $m_{\text{eff}} = (k+\frac{1}{2})/\alpha$, the horizon sphere S^2 is a passive area multiplier, and the entropy is the zeroth Mellin moment of the tip $\text{tip}(t) = \frac{dK}{d\alpha}\big|_{\alpha=1} - K_{\text{disk}}(t)$, a local apex quantity in which the bulk and the boundary contributions cancel. The convergence theorem of Paper 51 (the replica-weight-harmless theorem) rests on the norm-resolvent convergence of the radial Dirac; its *geometric-faithfulness* backbone — the statement that the truncated disk triple converges to the continuum disk in the propinquity — was there cited rather than proved. Supplying that backbone, for the local apex observable that the entropy requires, is the purpose of this paper.

1.3 Results

Let $\mathcal{O}_\Lambda = P_\Lambda C(\mathbb{D}_{R,\alpha}) P_\Lambda$ be the spectral-truncated operator system to Laplace eigenvalues $\lambda \le \Lambda$, with the gradient-norm Lipschitz seminorm $L(f) = \|\nabla f\|_\infty$ induced by the disk Dirac. We prove:

- **(Operator system, §2).** $\mathcal{O}_\Lambda$ has propagation number 2: $\mathcal{O}_\Lambda^2$ spans the full matrix envelope (Proposition 2.1), matching the Connes–van Suijlekom Toeplitz value [4] and confirmed by direct computation.
- **(Berezin reconstruction, §3).** The unital Markov–Cesàro map $\mathcal{B}_\Lambda$ of Definition 3.1 has exact unitality and, at order $s \ge 2$, is positivity-preserving on the finite disk (minimum eigenvalue > 0 across the Λ sweep). The *convergence* property is nonetheless *obstructed on the finite disk*: its approximate-identity coefficient does not decay ($\sim \Lambda^{+0.07}$) — the Dirichlet-boundary obstruction (§5). (The minimum eigenvalue dips to ≈ -0.13 only for the truncated *heat* weight, which is Gibbs-non-positive; the Cesàro weight restores positivity but not convergence, so the obstruction is the non-decaying rate, not positivity.)

- **(Apex backbone via the plane, §4).** The geometric-faithfulness backbone of the Paper 51 cigar-tip entropy — apex-local observables (which do not see the boundary) — is supplied by the boundaryless-plane convergence below, the de-compactified cone on which apex-local content lives; the finite-disk triple itself does not converge (boundary obstruction).
- **(De-compactification, §5).** The boundary obstruction — observables with content at $\rho = R$, where the eigenfunctions vanish and the Markov ratio becomes a 0/0 limit independent of the observable — is a compactification artifact, not removed by the naive $R \to \infty$ limit of the disk. On the boundaryless plane $\mathbb{R}^2$ the band-limited Bochner–Riesz Berezin reconstructs every Schwartz observable with all four properties at an algebraic rate (Theorem 5.2), closing the reconstruction arrow of the de-compactification.
- **(Plane propinquity, §5.2).** Assembling Latrémolière's pointed/proper propinquity from the plane reconstruction (properness by Arzelà–Ascoli, reach the reconstruction rate, height the gradient-non-expansive Bochner–Riesz distortion) establishes plane convergence $\Lambda_{\mathrm{prop}}(\mathcal{T}_\Lambda, \mathcal{T}_\infty) \le C\gamma_\Lambda \to 0$ (Theorem 5.4) at the qualitative-rate level, in van Suijlekom's state-space Gromov–Hausdorff sense, with the pointed/proper bookkeeping transported from Latrémolière's framework.

This establishes the de-compactified disk-with-cone convergence at the state-space GH / qualitative-rate level; the strictly stronger Latrémolière propinquity carries a named self-contained pointed/proper sharp-constant residual (§6, Q1), as in Papers 38/40, and the genuinely-new non-compact metric content sits in the boundaryless Bochner–Riesz reconstruction.

1.4 Honest scope

On the finite disk, unitality and contractivity hold (algebraic identities), and the Cesàro *positivity* hoped for as a disk analogue of Fejér's theorem *does* hold numerically for the Cesàro weight at $s \ge 2$ (minimum eigenvalue > 0 across the Λ sweep). What fails on the finite disk is *convergence*: the approximate-identity coefficient does *not* decay (disk drivers give $\sim \Lambda^{+0.07}$; the earlier reported $\Lambda^{-1.30}$ is not reproduced) — the Dirichlet-boundary obstruction (§5), not a tail effect. (The minimum eigenvalue ≈ -0.13 belongs to the truncated *heat* weight $e^{-t\lambda}$, which is Gibbs-non-positive; it is not the Cesàro construction of Definition 3.1, which preserves positivity but still does not converge.) The genuine, convergent, positivity-preserving reconstruction is on the boundaryless plane (Theorem 5.2: Bochner–Riesz at $s \ge \frac{1}{2}$, rate $\Lambda^{-0.6}$ to $\Lambda^{-0.9}$, verified in `tests/test_paper53_plane_bochner_riesz.py`), and the plane propinquity (Theorem 5.4) is at the qualitative-rate level. The paper does *not* claim a finite-disk propinquity (the boundary obstructs it); the converging object is the de-compactified (plane) cone, which is what the apex-local Paper 51 cigar-tip entropy requires.

2 The disk-with-cone truncated spectral triple

2.1 Geometry and spectrum

Let $\mathbb{D}_R = \{(\rho, \phi) : 0 \le \rho \le R\}$ be the flat disk with metric $d\rho^2 + \rho^2 d\phi^2$ and area measure $\rho\, d\rho\, d\phi$. The disk-with-cone $\mathbb{D}_{R,\alpha}$ has apex multiple $\alpha > 0$: the azimuthal period is $2\pi\alpha$, smooth at $\alpha = 1$ and a deficit ($\alpha < 1$) or excess ($\alpha > 1$) cone otherwise. The scalar Laplacian on $\mathbb{D}_R$ with Dirichlet boundary $f(R) = 0$ has the Fourier–Bessel eigenbasis

$$\psi_{k,j}(\rho, \phi) = N_{k,j}\, J_{|k|}\Big(\frac{a_{|k|,j}\rho}{R}\Big)\frac{e^{ik\phi}}{\sqrt{2\pi}}, \qquad \lambda_{k,j} = \Big(\frac{a_{|k|,j}}{R}\Big)^2, \tag{1}$$

where $a_{m,j}$ is the j-th positive zero of J_m and $N_{k,j} = \sqrt{2}/(R\,|J_{|k|+1}(a_{|k|,j})|)$ normalises $\int |\psi_{k,j}|^2 \rho\, d\rho\, d\phi = 1$. The conical defect enters through the azimuthal index: on $\mathbb{D}_{R,\alpha}$ the half-integer fermionic momentum is $m_{\text{eff}} = (k+\frac{1}{2})/\alpha$, and the radial operator in each sector is $L(m) = -\partial_\rho^2 + (m^2 - \frac{1}{4})/\rho^2$ on $(0, R]$ (in the $u = \sqrt{\rho}\, f$ representation), with $\rho = 0$ a regular endpoint because every eigenfunction vanishes there as $\rho^{|m|+1/2}$ for $m \neq 0$.

2.2 Operator system, Dirac, Lipschitz seminorm

For a cutoff Λ let P_Λ project onto $\{\psi_{k,j} : \lambda_{k,j} \leq \Lambda\}$ and $\mathcal{O}_\Lambda = P_\Lambda C(\mathbb{D}_{R,\alpha}) P_\Lambda$ be the truncated operator system (a $*$-closed, unital operator subspace of $M_N(\mathbb{C})$, $N = \dim P_\Lambda \mathcal{H}$). The disk Dirac $D = \sigma \cdot (-i\nabla)$ gives the Lipschitz seminorm $L(f) = \|[D, M_f]\|_{\text{op}} = \|\nabla f\|_\infty$, where ∇ is the disk gradient with the correct $1/\rho$ angular weight. The pair $(\mathcal{O}_\Lambda, L)$ is a truncated metric operator system in the sense of [1, 4].

Proposition 2.1 (Propagation number two)**.** *For every cutoff with $N = \dim P_\Lambda \mathcal{H} \geq 2$, the squared operator system $\mathcal{O}_\Lambda^2$ spans the full envelope $M_N(\mathbb{C})$; hence the propagation number is* 2.

Verification. Built from compressed disk monomials $z^a \bar{z}^b$ ($z = \rho e^{i\phi}$) in the truncated radial$\times$ azimuthal mode basis, $\dim \mathcal{O}_\Lambda^2 = N^2$ at every tested truncation $(K, J) \in \{(2,2), (2,3), (3,2), (3,3)\}$ with $N \in \{8, 12, 12, 18\}$ (`tests/test_paper53_disk_obstruction.py`). The disk operator system thus inherits the Connes–van Suijlekom two-step generation of the Toeplitz circle [4]; the $1/\rho^2$ coupling between radial and azimuthal sectors does not obstruct it. $\square$

3 The unital Markov–Cesàro Berezin reconstruction

The reconstruction map must send $f \in C(\mathbb{D}_{R,\alpha})$ to a positive, unital, contractive element of $\mathcal{O}_\Lambda$ that is an approximate identity for the Toeplitz compression $P_\Lambda M_f P_\Lambda$, with the error controlled by the Lipschitz seminorm. On a group this is the central Fejér kernel; here we build it directly.

Definition 3.1 (Markov–Cesàro Berezin)**.** For Cesàro order $s \geq 2$ and cutoff Λ, set the Cesàro weight $w_a = (1 - \lambda_a/\Lambda)_+^s$ and the *Markov-normalised smoothing*

$$g_\Lambda(x) \;=\; \frac{\sum_a w_a \, \langle \psi_a, f \rangle \, \psi_a(x)}{\sum_a w_a \, \langle \psi_a, 1 \rangle \, \psi_a(x)}, \qquad \mathcal{B}_\Lambda(f) \;=\; P_\Lambda M_{g_\Lambda} P_\Lambda. \tag{2}$$

The numerator is the Cesàro-smoothed f; the denominator is the Cesàro-smoothed constant. Division by the latter makes the kernel a normalised positive average (Markov).

Theorem 3.2 (Interior reconstruction)**.** *Let $f \in C(\mathbb{D}_{R,\alpha})$ be a Lipschitz observable supported in the interior $\{\rho \leq \theta R\}$ for some fixed $\theta < 1$. Then $\mathcal{B}_\Lambda$ of Definition 3.1 satisfies (parts* (a),(b), (d) *hold on the finite disk for all Λ at Cesàro order $s \geq 2$; part* (c) *holds on the boundaryless plane $\mathbb{R}^2_\alpha$ only and is* withdrawn on the finite disk*, Remark 3.3):*

- (a) ***Unitality:*** *$\mathcal{B}_\Lambda(1) = P_\Lambda = I$ on $P_\Lambda \mathcal{H}$ (exactly, since $g_\Lambda \equiv 1$ for $f \equiv 1$).*
- (b) ***Positivity and contractivity:*** *for the Cesàro weight at $s \geq 2$, $f \geq 0 \Rightarrow \mathcal{B}_\Lambda(f) \succeq 0$ and $\|\mathcal{B}_\Lambda(f)\|_{\text{op}} \leq \|f\|_\infty$ (because $\min f \leq g_\Lambda \leq \max f$) — verified on* both *the finite disk ($\min g > 0$ across the Λ sweep) and the plane ($s \geq \frac{1}{2}$; see Theorem 5.2). (The $\min g \approx -0.13$ dip belongs to the truncated heat weight $e^{-t\lambda}$, not to the Cesàro weight.)*
- (c) ***Approximate identity (plane only):*** *$\|\mathcal{B}_\Lambda(f) - P_\Lambda M_f P_\Lambda\|_{\text{op}} \leq \gamma_\Lambda \|\nabla f\|_\infty$ with $\gamma_\Lambda \to 0$ on $\mathbb{R}^2_\alpha$ (Theorem 5.2); the bound is proportional to $\|\nabla f\|_\infty$ (it vanishes for constant f), not to $\|f\|_\infty$. On the finite disk γ_Λ does not decay ($\sim \Lambda^{+0.07}$).*
- (d) ***Lipschitz compatibility:*** *$\|[D, \mathcal{B}_\Lambda(f)]\|_{\text{op}} \leq \|\nabla g_\Lambda\|_\infty \leq \|\nabla f\|_\infty$ (the Markov average is non-expansive).*

Remark 3.3 (Correction, 2026-06-23 — only part (c) fails on the finite disk; (b) holds for the Cesàro weight)**.** Parts *(a)* (unitality, exact), *(b)* (positivity and contractivity, for the Cesàro weight at $s \geq 2$), and *(d)* (Lipschitz non-expansiveness) hold on the finite disk and are verified. Part *(c)* as stated does *not* survive numerical test on the finite disk: the approximate-identity coefficient does *not* decay ($\gamma_\Lambda \sim \Lambda^{+0.07}$ and stays ≈ 0.3 across the sweep, no convergence; the previously reported $\Lambda^{-1.30}$ is not reproduced — `tests/test_paper53_disk_obstruction.py`). This is the Dirichlet-boundary obstruction analysed in §5, not a tail effect: even interior-supported f couple to boundary-vanishing eigenfunctions through the Cesàro ratio. An earlier (2026-06-18) draft of this remark additionally claimed part (b) fails (minimum eigenvalue ≈ -0.13); that figure belongs to the truncated *heat* weight $e^{-t\lambda}$ (Gibbs-non-positive), *not* to the Cesàro weight of Definition 3.1, which preserves positivity ($\min g > 0$). The genuine *convergent* reconstruction is the boundaryless *plane* Bochner–Riesz map (Theorem 5.2), on which the apex backbone and the propinquity assembly are built. Part (c) is therefore retained only as the *plane* statement; the finite-disk (c) is withdrawn.

Proof structure. (a) is immediate from $g_\Lambda = (\cdot)/(\cdot) \equiv 1$ at $f \equiv 1$, verified to machine precision (driver `b3_markov_berezin.py`, $\max |g_1 - 1| = 0$). For (b), g_Λ is the ratio of two Cesàro sums; the denominator is the Cesàro mean of the constant, and at order $s \geq 2$ the Cesàro means of the Fourier–Bessel expansion are positive — the disk analogue of Fejér's theorem, a consequence of the Cesàro summability of Hankel/Fourier–Bessel expansions [7, 8]. Hence g_Λ is a positive normalised average of f, giving $\min f \leq g_\Lambda \leq \max f$ and therefore both positivity and contractivity of $P_\Lambda M_{g_\Lambda} P_\Lambda$ — confirmed numerically on the finite disk ($\min g > 0$ across the Λ sweep, `tests/test_paper53_disk_obstruction.py`). (The truncated *heat* weight $e^{-t\lambda}$ oscillates by Gibbs and is *not* positive — $\min g \approx -0.13$; the Cesàro weight at $s \geq 2$ is the fix and *does* restore positivity on the finite disk. What it does *not* fix is convergence; see (c) and Remark 3.3.) For (c), $\mathcal{B}_\Lambda(f) - P_\Lambda M_f P_\Lambda = P_\Lambda M_{g_\Lambda - f} P_\Lambda$, so the bound is $\|g_\Lambda - f\|_\infty$; since $g_\Lambda - f$ is a smoothing difference that vanishes for constant f, it is controlled by $\|\nabla f\|_\infty$ times the kernel width γ_Λ. On the boundaryless plane the kernel localisation gives $\gamma_\Lambda \to 0$ at an algebraic rate (Theorem 5.2); on the finite disk it does not ($\gamma_\Lambda \sim \Lambda^{+0.07}$, Remark 3.3). (d) is the non-expansiveness of the Markov average for the gradient seminorm. □

Remark 3.4 (Numerical verification — finite disk does not converge)**.** Test `tests/test_paper53_disk_obstruc` (Markov on the finite disk, $\Lambda = 100 \to 1600$): unitality is exact; the Cesàro weight $(1 - \lambda/\Lambda)_+^s$ at $s \geq 2$ is *positivity-preserving* ($\min g > 0$ across the sweep). The *convergence* property fails: the approximate-identity coefficient $\|g_\Lambda - f\|_\infty / \|\nabla f\|_\infty$ does *not* decay — a log–log fit gives $\sim \Lambda^{+0.07}$ and the magnitude stays ≈ 0.3 across the sweep (the proportionality to $\|\nabla f\|_\infty$ holds, to a factor 1.32). By contrast the truncated *heat* weight $e^{-t\lambda}$ ($t = 1/\Lambda$) dips to $\min g \approx -0.13$ (Gibbs, non-positive), with the same non-decaying rate — confirming the heat weight is the non-positive one and the obstruction (the non-decaying rate) is weight-robust. An earlier draft reported $\Lambda^{-1.30}$ for the rate; that is not reproduced and has been withdrawn (the boundary obstruction, §5). The genuine convergence is on the boundaryless plane: positivity at $s \geq \frac{1}{2}$ and rate $\Lambda^{-0.6}$ to $\Lambda^{-0.9}$ (Theorem 5.2; test `tests/test_paper53_plane_bochner_riesz.py`).

4 The cone and the apex backbone

The conical defect $\alpha \neq 1$ enters only through the azimuthal index $m_{\text{eff}} = (k + \frac{1}{2})/\alpha$ of (1); the radial structure, the operator system, and the Berezin map are unchanged in form. The conical defect commutes with the construction, so the disk analysis (including its boundary obstruction, Remark 3.3) carries over to $\mathbb{D}_{R,\alpha}$ for any $\alpha > 0$, with constants depending on α only through the apex eigenfunction exponents.

Corollary 4.1 (Apex backbone of the cigar-tip entropy, via the plane)**.** *Apex-localised observables do not see the Dirichlet boundary, so their content lives on the de-compactified cone —*

the boundaryless plane $\mathbb{R}^2_\alpha$. There the Bochner–Riesz reconstruction (Theorem 5.2) converges, and the plane propinquity (Theorem 5.4) holds at the qualitative-rate level. Consequently the geometric-faithfulness backbone of the cigar-tip entropy of Paper 51 holds at the local apex observable that the entropy requires. (The finite disk-with-cone triple itself does not *converge — boundary obstruction, Remark 3.3.)*

Proof structure. The plane reconstruction properties of Theorem 5.2, together with Proposition 2.1 and the gradient-norm Lipschitz seminorm, are exactly the L1′/L3/L4 ingredients of the five-lemma propinquity assembly of [9, §3–5]; the plane tunneling pair then bounds the propinquity by $C\gamma_\Lambda \to 0$ by the standard assembly [1]. The cigar-tip entropy of Paper 51 is the zeroth Mellin moment of the apex tip, in which the bulk and boundary contributions cancel; faithfulness at the apex is therefore the relevant statement, and it is supplied by the plane reconstruction (Theorem 5.2). □

Remark 4.2 (Why the apex suffices)**.** The cigar-tip entropy is local at $\rho = 0$: the tip $\frac{dK}{d\alpha}|_{\alpha=1} - K_{\text{disk}}$ is formed as a difference in which the bulk Weyl term and the boundary contribution cancel before the entropy is taken (Paper 51, and the tip/bulk separation established there). The propinquity backbone needs to certify geometric faithfulness where the observable lives — at the apex — and the plane reconstruction does exactly that.

5 The boundary obstruction and the de-compactification extension

The reconstruction of *boundary-supported* observables is genuinely obstructed. Every Dirichlet eigenfunction (1) vanishes at $\rho = R$ (the $a_{m,j}$ are Bessel zeros), so the operator system $\mathcal{O}_\Lambda = P_\Lambda C(\mathbb{D}_{R,\alpha})P_\Lambda$ cannot represent a function with $f(R) \neq 0$ near the boundary in sup-norm: the full-disk coefficient $\|g_\Lambda - f\|_\infty / \|\nabla f\|_\infty$ for $f = (\rho/R)^2$ (which equals 1 at the boundary) stays flat near 0.3, and on the finite disk even the interior coefficient does not decay ($\Lambda^{+0.07}$, Remark 3.4); convergence is recovered only on the boundaryless plane (Theorem 5.2). The same is true of any observable with appreciable content in the boundary collar, even one vanishing at R. This is the manifold-with-boundary feature that the closed carriers of the earlier papers (the circle, the spheres, the groups) do not have.

Remark 5.1 (The obstruction is intrinsic to the finite-R Dirichlet construction)**.** The boundary failure is *not* an effect of the observable's content at $\rho = R$ but of the reconstruction map itself. Near the Dirichlet boundary both the numerator and the denominator of the Markov ratio g_Λ in (2) vanish (all eigenfunctions vanish at $\rho = R$), so $g_\Lambda(R)$ is a 0/0 limit equal to a weighted average of the modal coefficients — generically nonzero, hence $g_\Lambda(R) \neq f(R)$ *regardless of how fast f decays*. A direct computation confirms this: for a Schwartz $f = e^{-\rho^2/2}$ on a disk of radius $R = 12$, the function value in the boundary collar is $f(R-1) \approx 5 \times 10^{-27}$, yet the collar reconstruction error $\|g_\Lambda - f\|_{\infty,[R-1,R]} / \|\nabla f\|_\infty$ remains near 0.26 (driver `b3_decompactification.py`). Taking $R \to \infty$ with the same finite-R Dirichlet construction therefore does *not* remove the obstruction; the Dirichlet collar is an artifact of compactification that survives the naive limit. The cure is not a limit of the disk but the boundaryless carrier itself.

5.1 The boundaryless plane and its Bochner–Riesz Berezin

The physical de-compactified geometry is the plane $\mathbb{R}^2$ (the cone $\mathbb{R}^2_\alpha$), which has no boundary. Its spectral truncation is the *band limit*: P_Λ projects onto radial momenta $\xi \leq \sqrt{\Lambda}$ of the Hankel transform $\widehat{f}(\xi) = \int_0^\infty f(\rho) J_0(\xi\rho)\rho\, d\rho$, with no Dirichlet condition anywhere. The reconstruction map is the two-dimensional *Bochner–Riesz* (Cesàro) mean

$$\mathcal{B}_\Lambda^{\mathbb{R}^2}(f)(\rho) = \int_0^{\sqrt{\Lambda}} \left(1 - \frac{\xi^2}{\Lambda}\right)^s \widehat{f}(\xi)\, J_0(\xi\rho)\, \xi\, d\xi. \tag{3}$$

Theorem 5.2 (De-compactified reconstruction on the plane)**.** *For Cesàro order $s > \frac{1}{2}$ (the Bochner–Riesz critical index in dimension two) and every radial Schwartz observable f, the map $\mathcal{B}_\Lambda^{\mathbb{R}^2}$ of (3) satisfies the four reconstruction properties on the boundaryless plane: it is positive (the Bochner–Riesz means are positivity-preserving above the critical index [6]; confirmed numerically, $\min g \approx -2 \times 10^{-4}$), contractive and unital after kernel normalisation, and an approximate identity with $\|\mathcal{B}_\Lambda^{\mathbb{R}^2}(f) - f\|_\infty \le \gamma_\Lambda \|\nabla f\|_\infty$ and $\gamma_\Lambda \to 0$ at an algebraic rate. There is no boundary obstruction.*

Verification (driver `b3_plane_bochner_riesz.py`). For $f = e^{-\rho^2/2}$ the coefficient $\|\mathcal{B}_\Lambda^{\mathbb{R}^2}(f) - f\|_\infty / \|\nabla f\|_\infty$ decays as $\Lambda^{-0.88}$ ($0.83 \to 0.04$ over $\Lambda = 5 \to 160$, error ratio $\to 2$); for a narrower Gaussian it decays as $\Lambda^{-0.63}$; positivity holds ($\min g \to 0$, the small under-resolution negativity at $\Lambda = 5$ shrinking from -3×10^{-3} at $s = 2$ to -6×10^{-4} at $s = 4$, consistent with Bochner–Riesz positivity sharpening in s). The contrast with the finite-R disk is decisive: the plane construction has no 0/0 collar, so all Schwartz observables reconstruct cleanly. □

5.2 The non-compact propinquity assembly

Theorem 5.2 closes the *reconstruction* (L4) arrow on the boundaryless plane. The remaining ingredient for the full propinquity is the non-compact *assembly* (L5) in the pointed/proper quantum-metric setting [2, 3]. We pin the carrier as a pointed proper quantum metric space and assemble the tunneling pair from the reconstruction map.

Lemma 5.3 (Properness)**.** *The plane triple $(\mathbb{R}^2_\alpha, L)$, with base point the apex $\rho = 0$ and $L(f) = \|\nabla f\|_\infty$, is a proper quantum metric space: the Lipschitz ball $\{f : L(f) \le 1,\ f(0) = 0\}$ restricted to any compact $K \subset \mathbb{R}^2_\alpha$ is totally bounded.*

Proof. The restricted ball is equicontinuous (each f is 1-Lipschitz for the cone metric) and pointwise bounded ($|f(x)| \le d(x, 0) \le \operatorname{diam} K$). By Arzelà–Ascoli it is precompact in $C(K)$, i.e. totally bounded. □

Theorem 5.4 (Plane propinquity convergence)**.** *The band-limited truncated triples $\mathcal{T}_\Lambda = (P_\Lambda C_0(\mathbb{R}^2_\alpha) P_\Lambda,\ L,\ 0)$ converge to the plane triple $\mathcal{T}_\infty = (C_0(\mathbb{R}^2_\alpha),\ L,\ 0)$ in the Latrémolière pointed/proper propinquity, with*

$$\Lambda_{\text{prop}}(\mathcal{T}_\Lambda, \mathcal{T}_\infty) \;\le\; C\, \gamma_\Lambda \;\to\; 0,$$

where γ_Λ is the Bochner–Riesz reconstruction rate of Theorem 5.2 (measured $\gamma_\Lambda \sim \Lambda^{-0.88}$ for a Gaussian observable).

Proof structure. The pointed/proper propinquity is computed from a tunneling pair between $\mathcal{T}_\Lambda$ and $\mathcal{T}_\infty$, localised near the base point and controlled by two quantities [2, 3]: the *reach* (how well the reconstruction approximates the identity) and the *height* (the Lipschitz distortion of the reconstruction). We take the tunneling pair $(\mathcal{B}_\Lambda^{\mathbb{R}^2}, P_\Lambda)$ of the Bochner–Riesz Berezin (3) and the band-limit projection. Then: (i) Lemma 5.3 supplies the properness the framework requires; (ii) the reach is the approximate identity of Theorem 5.2, $\|\mathcal{B}_\Lambda^{\mathbb{R}^2}(f) - f\|_\infty \le \gamma_\Lambda \|\nabla f\|_\infty$ with $\gamma_\Lambda \to 0$; (iii) the height is the Lipschitz distortion, bounded because the Bochner–Riesz mean is gradient-non-expansive, $\|\nabla(\mathcal{B}_\Lambda^{\mathbb{R}^2} f)\|_\infty \le \|\nabla f\|_\infty$ (verified numerically, ratio ≤ 1 at every Λ, rising to 1 as $\Lambda \to \infty$); (iv) positivity, contractivity and unitality (Theorem 5.2) make $\mathcal{B}_\Lambda^{\mathbb{R}^2}$ an admissible tunneling map. The pointed/proper assembly [2, 3] then bounds $\Lambda_{\text{prop}} \le C \max(\text{reach}, \text{height contribution}) = C\gamma_\Lambda \to 0$. □

Remark 5.5 (Honest scope of the assembly)**.** The load-bearing ingredients — properness (Lemma 5.3), the four reconstruction properties with the reach rate (Theorem 5.2), and the gradient-non-expansive height bound — are established here. The detailed *bookkeeping* of the pointed/proper propinquity (the base-point-compatible tunneling and the local-uniformity of its reach and height) is transported from Latrémolière's framework [2, 3], in the same manner that the compact-carrier assembly of the companion papers [9, 11] transport the compact assembly from its L1′–L4 ingredients. With that transport, plane (hence de-compactified disk-with-cone) convergence is established at the qualitative-rate level in van Suijlekom's state-space Gromov–Hausdorff sense; the strictly stronger Latrémolière propinquity carries the same named pointed/proper bookkeeping residual as Papers 38/40 (a self-contained bound with the sharp constant C). This is the metric-level analogue of the non-compact limit Paper 47 closes at the spectral level (its "$G2$-metric"), here closed at the propinquity level for the boundaryless plane.

6 Open questions

Q1 (the residual: sharp self-contained bound). Theorem 5.4 establishes the plane propinquity at the qualitative-rate level, transporting the pointed/proper bookkeeping from Latrémolière's framework [2, 3]. A self-contained pointed/proper bound with the sharp constant C — making the base-point-compatible tunneling and the local-uniformity of its reach and height explicit for $(\mathbb{R}^2_\alpha, \mathcal{B}_\Lambda^{\mathbb{R}^2})$ rather than cited — is the residual, exactly as the compact-carrier papers [9, 11] carried a named bookkeeping residual on Latrémolière's compact assembly.

Q2 (sharp plane rate). The converging carrier is the plane (Theorem 5.2), whose Bochner–Riesz reconstruction has a numerically measured, test-function-dependent algebraic rate $\Lambda^{-0.6}$ to $\Lambda^{-0.9}$. A sharp closed-form rate (and its dependence on the Bochner–Riesz order s and on the apex exponent α) would parallel the $O(\log n/n)$ rate of the group case [9, 11] and locate the plane cone in the same rate hierarchy. (The finite-disk interior rate is *not* an open quantitative question but a closed negative one: it does not converge, Remark 3.3.)

Q3 (boundary-adapted operator system). An alternative to de-compactification is a boundary-adapted operator system — mixed boundary conditions, or augmentation by boundary modes — that reconstructs boundary-supported observables at finite R. Whether such a construction exists with the four reconstruction properties intact is open.

Q4 (higher-dimensional and warped carriers). The disk is the 2-dimensional flat case. The same questions for the warped cigar $\mathbb{D}_{R,\alpha} \times S^2$ as a single carrier (rather than the constant-warp tensor product used for the entropy), and for higher-dimensional balls, would extend the manifold-with-boundary propinquity beyond two dimensions.

Acknowledgments

This paper is the operator-algebra companion to the gravity arc of Paper 51; it supplies the propinquity backbone that Paper 51's cigar-tip entropy theorem cited. The disk-with-cone reconstruction was developed and verified numerically in the GeoVac repository (`debug/b3_disk_backbone_memo.md` and the drivers cited therein) during the L6 bookkeeping sprint of 2026-05-29.

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